

Revision Anterior Cruciate Ligament Injuries

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As the number of primary anterior cruciate ligament (ACL) reconstructions performed annually continues to increase, the number of revision procedures is also projected to increase.^{1,2} A larger number of patients returning to high-demand sports and activities following primary reconstruction has resulted in an increased number of revision and repeat revision reconstructions.^{3,4} Failure of primary ACL reconstruction, defined by recurrent laxity or graft failure, has been reported in 2.9% to 44% of patients, with higher failure rates in younger patients and following allograft reconstruction.⁵⁻¹⁵ It is estimated that between 1.7% and 7.7% of patients who underwent primary ACL reconstruction will undergo revision reconstruction.^{1,16-20} As approximately 175,000 to 200,000 primary reconstruction procedures are performed in the United States annually, it is estimated that approximately 3000 to 13,000 patients will undergo revision ACL reconstruction each year. This group is typically composed of a young, active population that desires a return to their previous activities.^{21,22}

Revision ACL reconstruction is a challenging procedure with technical issues that can include retained hardware, incorrect bone tunnel placement and dilation, and limited graft choices.^{3,23-26} This procedure requires proficiency with multiple techniques and flexibility when encountering unexpected obstacles to a successful revision reconstruction. Furthermore, patients undergoing revision ACL reconstruction have a lower rate of return to sport, an increased incidence of meniscal and chondral injuries, and inferior clinical outcomes compared with primary reconstruction.^{8,9,11,21,27-29} Although excellent results can be achieved following a well-executed revision ACL reconstruction, patients must be counseled regarding expectations following this procedure.

Epidemiology

Recurrent instability is defined as failure of the reconstructed ligament to provide adequate anterior and rotatory stability to the knee. Recurrent instability or graft failure necessitating revision most commonly affects patients in the third decade of life.^{8,22,30,31} Risk factors for graft failure include male gender, return to sports involving pivoting or jumping, contact sports, allograft, and age younger than 25 years.^{14,21,29,32-35} Although patients in their 20s comprise the highest absolute number of revision ACL reconstructions, those aged 10 to 19 have the highest incidence of graft failure, and each 10-year decrease in age has been shown to increase the odds of graft failure by 2.3-fold.¹⁴ This is likely attributable to a higher activity level among younger patients

with ACL injuries, as well as a higher likelihood of premature return to activities prior to adequate rehabilitation.³⁵ Although females have a higher rate of tearing an intact contralateral ACL, males are more likely to tear the reconstructed graft within the first 2 years following ACL reconstruction²¹; 55% to 70% of revision reconstructions are performed in male patients.^{11,21,22,33,36-39} Although the reasons for higher rates of revision reconstructions among males have not been elucidated, males may have an increased propensity to return to high-risk activities that put the graft at risk.

Approximately 90% of revision reconstructions are first-time revision procedures.²² Seventy percent of patients report a non-contact injury resulting in rerupture, with 40% occurring during cutting and 30% during jumping activities. Three-quarters of patients report an injury while playing sports, with most occurring during soccer and basketball. The mean time from prior reconstruction to revision is greater than 2 years in approximately 66% of patients, 1 to 2 years in 20% of patients, and less than 1 year in 15% of patients.²² Of all patients undergoing revision ACL reconstruction, autograft was found to be used in the prior reconstruction in approximately 70% of cases and allograft in 30% of cases.²²

HISTORY

ACL graft failure should be considered in the setting of objective sagittal (anteroposterior [AP]) or rotatory knee laxity, subjective knee instability, knee pain following prior ACL reconstruction, extensor mechanism dysfunction, and infection.²⁹ Knee pain should be distinguished from instability.²⁸ A complete history should be obtained, including mechanism of injury, quality of symptoms (swelling, giving way, locking, catching, crepitus, gait changes), symptom duration, previous injuries and surgical interventions including ligamentous, meniscal, and articular cartilage injuries, graft type and source, and graft fixation.^{28,29} If the patient describing recurrent instability is unable to recall a causative traumatic episode, this may suggest technical or biologic reasons for graft failure. The patient should be asked to describe the postoperative course following the previous reconstruction, detailing the time course to rehabilitation milestones and return to activity/sport. Failure to return to the same level of activity postoperatively may suggest a technical error or inadequate rehabilitation.²⁹ Inadequate postoperative rehabilitation resulting in poor proprioception, deconditioning, stiffness, and

Abstract

Revision anterior cruciate ligament (ACL) reconstruction is a challenging procedure with technical issues requiring proficiency with multiple techniques and flexibility when encountering unexpected obstacles to a successful revision reconstruction. Patients undergoing revision ACL reconstruction have a lower rate of return to sport, an increased incidence of meniscal and chondral injuries, and inferior clinical outcomes compared with primary reconstruction. As such, these patients must be counseled regarding expectations following this procedure.

In addition to a complete history and physical examination, preoperative evaluation should include imaging to assess tunnel positioning. Tunnel malpositioning is the most technical error during both primary and revision ACL reconstruction and must be addressed at the time of revision reconstruction. Untreated ligamentous injury and malalignment must also be addressed before or during revision reconstruction to help prevent graft failure. Similar to primary reconstruction, allograft has been shown to have higher failure rate than autograft in the revision setting.

Several factors must be considered in the setting of revision ACL reconstruction, including graft selection, tunnel locations, tunnel sizes and need for bone grafting, previous fixation type and need for removal, method of revision fixation, and postoperative protocol. The revision reconstruction often must be adapted to the technique previously used. We describe our preferred technique, using a rear-entry two-incision approach to facilitate femoral tunnel drilling independent of the tibial tunnel position. While the principles of postoperative management are similar to primary reconstruction, rehabilitation and return to sport is generally slower than primary reconstruction.

Keywords

revision ACL
revision ACL reconstruction
ACL failure
revision ACL graft choice
revision ACL technique
ACL outcomes

pain may be unimproved after a revision reconstruction.^{28,29,40,41} Such patients benefit from preoperative rehabilitation. Previous operative notes, clinic notes, therapy notes, imaging studies, and intraoperative arthroscopic images should be reviewed. Associated intra-articular injuries, including meniscal and chondral injuries, diagnosed at the time of the previous reconstruction must be considered in the setting of patient dissatisfaction following a previous reconstruction.⁴²

PHYSICAL EXAM

Physical exam begins with assessment of lower extremity alignment, gait, muscle tone, specifically evaluating the bulk of the vastus medialis obliquus (VMO), and the location of previous incisions.²⁹ Range of motion (ROM) should be measured, assessing for a flexion contracture or extensor lag, and prone heel height examination may identify a subtle flexion contracture not appreciated while supine.²⁹ The ACL should be evaluated with the Lachman test to determine anterior laxity, and a pivot shift exam should be attempted to determine rotatory instability, with both exams compared with the uninjured contralateral leg.^{40,43} However, approximately 32% of autograft-reconstructed knees may have positive findings on a Lachman test and 22% positive findings on the pivot-shift test, suggesting that postoperative laxity may exist in a large number of patients after reconstruction despite satisfactory subjective outcomes.²⁸ Conversely, despite normal findings on Lachman and pivot shift examination, some patients may describe the subjective perception of knee instability, with an inability to trust their knee during pivoting and twisting activities. In equivocal cases, the KT-1000/2000 arthrometer (MEDmetric, San Diego, CA) may be used to provide a more objective measurement of AP laxity.^{28,29,44} A greater than 3-mm side-to-side difference correlates with failure of the native ACL, and multiple studies have used this criterion to quantify failure of a reconstructed ACL.⁴⁵⁻⁵¹ Other studies have defined graft failure as a greater than 5 mm side-to-side difference.^{24,52-57}

In addition to these ACL-specific examination tests, a complete knee exam should be performed including examination of the posterior cruciate ligament (PCL), medial collateral ligament (MCL) and lateral collateral ligament (LCL), posterolateral corner, and menisci. As described later in greater detail, associated injury to other ligamentous and soft tissue structures may contribute to instability following ACL reconstruction.⁵⁸

IMAGING

A complete weight-bearing series of knee radiographs, including AP in extension, flexion posteroanterior (PA) (Rosenberg), lateral, and axial (sunrise or merchant) views should be obtained in the setting of pain or instability following ACL reconstruction (Fig. 99.1). If there is concern for coronal malalignment or instability, full-length standing AP radiographs should also be obtained. Radiographs should be used to assess tunnel location, tunnel expansion and associated bone loss, osteoarthritis (OA) progression, coronal or sagittal malalignment, and the presence of hardware or implants that may affect revision surgical planning. A full extension lateral with the ankle supported to allow hyperextension can assess tibial tunnel position in relation to Blumensaat line.

Radiographic Tunnel Positioning

A malpositioned femoral tunnel is typically either too anterior and/or too vertical, and a malpositioned tibial tunnel is typically too anterior. On a lateral radiograph, the tibial plateau and Blumensaat line may be divided from anterior to posterior into four equal quadrants.⁵⁹ The tibial tunnel should enter the joint in the posterior third of the second quadrant, and the femoral tunnel should enter in the most posterior quadrant.^{59,60} Taking 0% as the anterior and 100% as the posterior extent of Blumensaat line, a femoral tunnel more than 40% anteriorly along Blumensaat line is considered excessively anterior.⁶⁰ Femoral tunnels located at least 60% posteriorly along Blumensaat line and tibial tunnels at least 20% posteriorly along the tibial plateau have

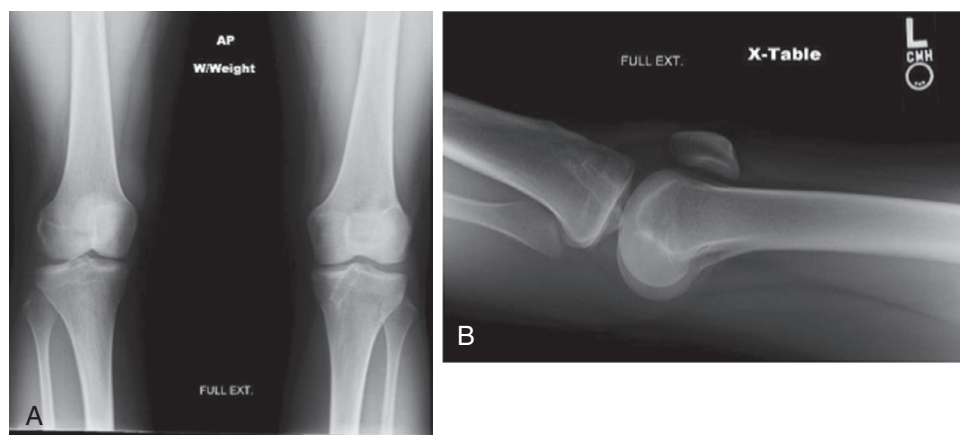


Fig. 99.1 Preoperative radiographs. (A) Standing anteroposterior radiographs demonstrate varus malalignment of the left knee in setting of prior anterior cruciate ligament (ACL) reconstruction. Full-length standing anteroposterior radiographs may be obtained in this setting to plan for concomitant high tibial osteotomy (HTO), which may be performed prior to or during revision ACL reconstruction. (B) Full extension lateral radiograph with ankle supported to allow hyperextension may be used to assess tibial tunnel position in relation to Blumensaat line. Notice knee hyperextension in setting of prior failed ACL reconstruction.

demonstrated improved clinical outcomes,⁶¹ whereas those with more anterior femoral or tibial tunnels are associated with increased failure rates and inferior clinical outcomes.⁶² Outcomes following revision for excessively anterior femoral tunnel malpositioning primary are improved compared to revision for another cause.⁶³

Femoral tunnel malpositioning in the coronal plane can also contribute to graft failure. On an AP or PA radiograph, fixation

along the anterior rather than lateral cortex may indicate a “vertical” femoral tunnel (Fig. 99.2).⁶⁴ A double-blinded study assessing the results of high and low femoral wall position showed improved International Knee Documentation Committee (IKDC) scores for the low position group.⁶⁵ The tibial tunnel should penetrate the articular surface at the midpoint of the plateau on an AP or PA radiograph. A cadaveric study using landmarks to determine anatomic tunnel location demonstrated the sagittal tibial

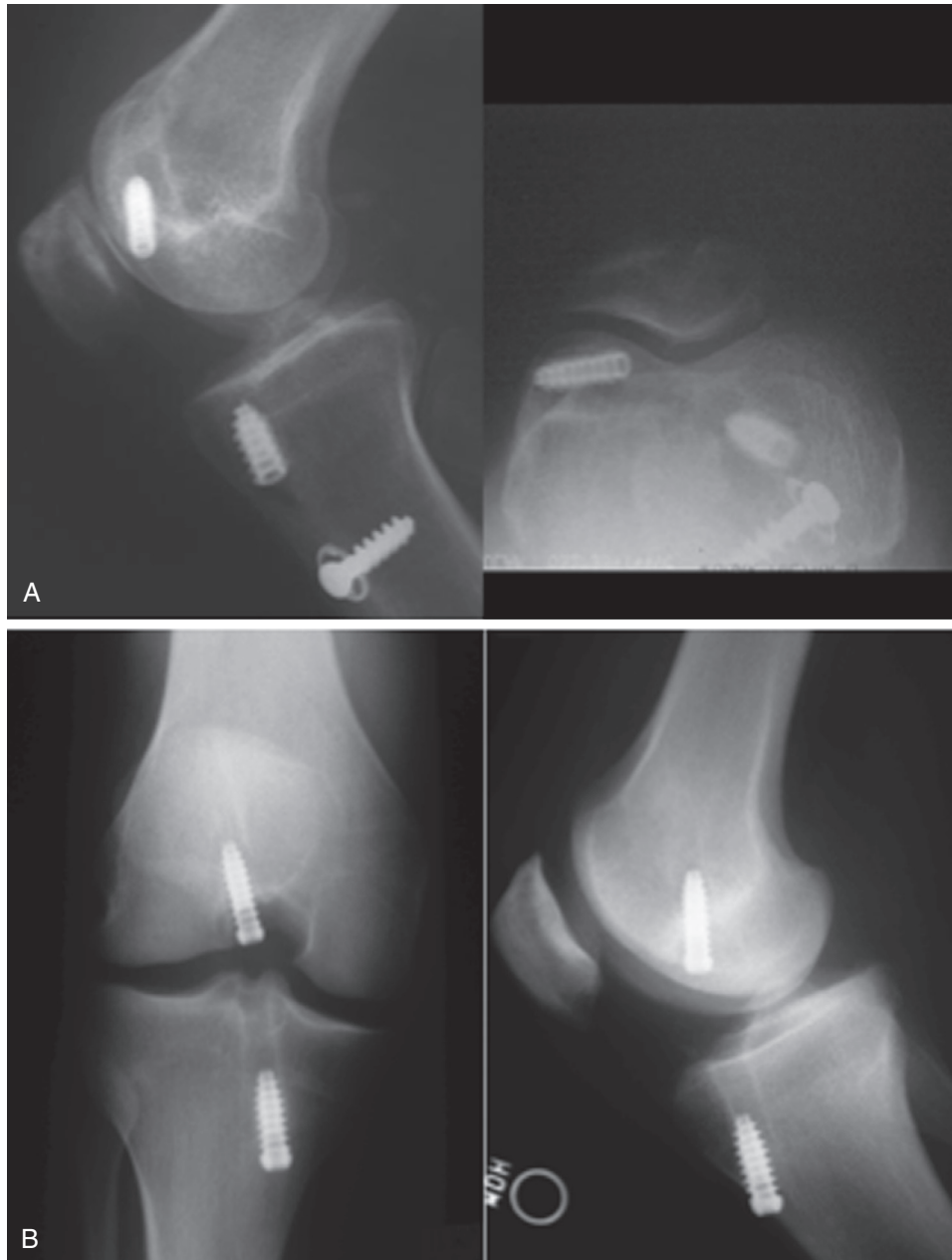


Fig. 99.2 Examples of incorrect tunnel placement. (A) Excessively anterior femoral and tibial tunnels. Taking 0% as the anterior and 100% as the posterior extent of Blumensaat line, a femoral tunnel more than 40% anteriorly along Blumensaat line is considered excessively anterior. The tibial tunnel should enter the joint in the posterior third of the second quadrant of the tibial plateau, and the femoral tunnel should enter in the most posterior quadrant of Blumensaat line. (B) Vertical femoral tunnel placement, best visualized on antero-posterior or posteroanterior (PA) radiograph. On a PA radiograph, fixation along the anterior rather than lateral cortex may indicate a “vertical” femoral tunnel. Lateral view also demonstrates anterior femoral and tibial tunnel placement.

tunnel angle to be 75 degrees and the coronal angle to be 65.7 degrees.⁶⁶ A tibial tunnel angle greater than 75 degrees in the coronal plane may result in a loss of flexion and increased sagittal (AP) laxity.^{74,75}

Graft Impingement

A hyperextension lateral radiograph is the best method to assess for anterior graft impingement,⁶⁷ which is associated with increased rates of effusion, lack of extension, and increased failure rates.⁶⁸⁻⁷¹ The Multicenter ACL Revision Study (MARS) group demonstrated that approximately 51% of patients undergoing revision reconstruction had no graft impingement, 47% had some degree of impingement, and 2% had complete impingement with the tibial tunnel completely anterior to Blumensaat line.⁶⁰ A study defined the distance posterior to the anterior edge of the tibia that minimizes the risk of anterior graft impingement,⁷² and if the center of the tibial tunnel at the articular surface was at least 22 to 28 mm posterior to the anterior edge of the tibia, then no graft impingement resulted.⁷³ Another study demonstrated that if the tibial tunnel is positioned in the posteromedial portion of the native ACL footprint, then graft impingement does not occur.⁶⁷

Advanced Imaging

Computed tomography (CT) may be obtained to provide a more detailed assessment of existing tunnel location, tunnel dilation, and bone quality (Fig. 99.3). Excessively posterior femoral tunnel placement may result in posterior wall blowout, limiting fixation options at the time of revision surgery to those relying on lateral cortical fixation or necessitating bone grafting followed by staged ACL reconstruction.^{76,77} The authors routinely obtain magnetic resonance imaging (MRI) preoperatively to assess for concomitant meniscal, chondral, and ligamentous injury, and it can also be useful to assess graft integrity and tunnel dilation. The authors routinely obtain MRI preoperatively. However, its utility may

be compromised by artifact if metallic implants were used for fixation during the prior reconstruction.⁷⁷

DECISION-MAKING PRINCIPLES

Causes of Failure of Anterior Cruciate Ligament Reconstruction

Instability following ACL reconstruction may be categorized into early (less than 6 months postoperative) or late (greater than 6 months postoperative) instability. Early laxity generally results from technical errors, failure of graft incorporation, loss of graft fixation, premature return to activity/sport, overly aggressive rehabilitation, or a combination of these factors.^{11,28,78-80} Late instability generally results from traumatic rerupture and less often from technical errors.³² A less common cause of late instability is failure to address concomitant ligamentous pathology at the time of primary reconstruction.^{77,81} With appropriate surgical technique and rehabilitation, primary ACL grafts are at no greater risk of rupture compared with the contralateral uninjured ACL.^{14,32,82}

Patients undergoing revision reconstructions report trauma resulting in recurrent instability in approximately 55% to 70% of cases, most often from a noncontact injury during sports.²² However, at the time of revision, the MARS surgeons have deemed the cause of failure to be traumatic in only one-third of cases, technical error in approximately one-quarter of cases, biologic failure (lack of graft incorporation) in 7% of cases, and multifactorial in 31%.²² Up to 53% of patients undergoing revision reconstruction have some degree of technical error contributing to graft failure,^{29,83} and among these patients, 80% have femoral tunnel malposition. Injury resulting in rupture of a well-positioned and well-fixed graft is nearly as frequent as failure due to incorrect femoral tunnel placement.^{22,33,55,63,84} Interestingly, two decades ago, failure due to malpositioned tunnels was two to three times more likely than traumatic rerupture of

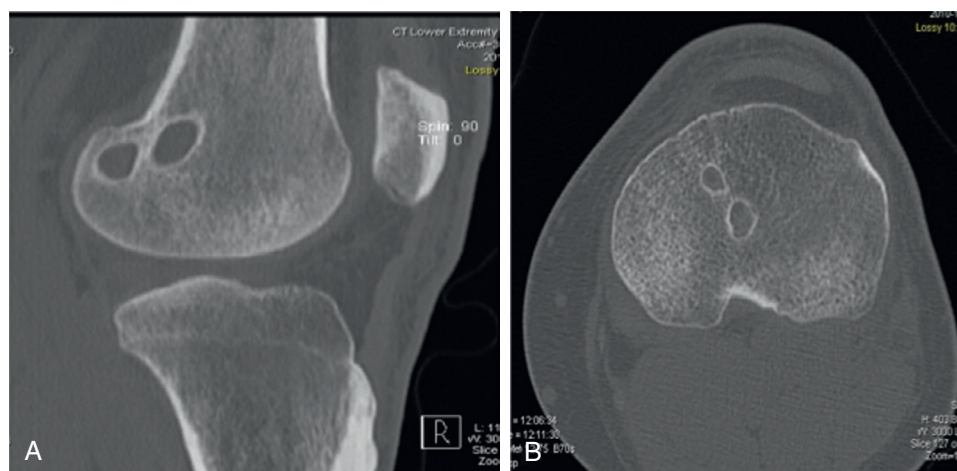


Fig. 99.3 Preoperative computed tomography scan may be obtained to assess tunnel location, size and bone quality. (A) Selected sagittal cut of lateral femoral condyle demonstrates two femoral tunnels occupying large portion of proximal lateral intercondylar notch. (B) Selected axial cut redemonstrates femoral tunnels. In this case a two-stage revision was performed in which these femoral tunnels were bone grafted prior to revision reconstruction. The large size and location of the existing femoral tunnels prevented the placement of a new, properly placed revision femoral tunnel.

a well-positioned, well-healed graft. These differences may be attributable to higher-demand patients expecting a return to their previous activity level, overaggressive rehabilitation protocols that may prevent adequate graft healing, and improved surgical techniques resulting in more consistently accurate tunnel placement during primary reconstruction.^{63,85,86}

Tunnel Malpositioning

Tunnel malpositioning is the most common technical error during primary and revision ACL reconstruction and can result in excessive graft forces and strain resulting in inadequate graft incorporation, graft loosening, and atraumatic graft failure.^{21,63,84,87,88} Occurring three times more frequently than tibial tunnel malpositioning, excessively anterior placement of the femoral tunnel is the most frequent technical error.^{29,84} This is, in part, due to difficulties in visualizing the native femoral footprint^{89,90} and results in a short graft with excessive graft tension in flexion and an initial loss of knee flexion.^{63,84} During rehabilitation and return to sport, recurrent stretching of the graft leads to laxity and eventual failure.^{29,63,84,91} Conversely, excessively posterior femoral tunnel placement results in excessive graft tension in extension with laxity in flexion.⁸⁴ A vertical femoral tunnel may provide sagittal (AP) stability but will result in rotational instability.⁹²

An excessively anterior tibial tunnel results in graft impingement against the intercondylar notch with loss of terminal extension.^{69,70,93-95} Excessively posterior tunnel placement results in graft impingement against the PCL and a vertical graft. This initially results in a loss of terminal flexion, with eventual graft laxity if full flexion is achieved.^{84,96} Errant placement of a tibial tunnel too far medially or laterally can also result in graft impingement on the intercondylar notch, potentially causing articular cartilage injury.⁹⁷

Untreated Ligamentous Laxity

Of patients undergoing revision ACL reconstruction, 3% to 31% of patients may have had unrecognized collateral ligament instability or malalignment contributing to graft failure.^{6,24,29,33,49,51-54,56,57,61,91,98-100} Untreated varus malalignment can result in varus thrust and graft attenuation.⁸¹ Similarly, untreated posterolateral or posteromedial corner injuries may result in excessive forces within the graft and early failure; these injuries should be addressed before or during revision ACL reconstruction.^{25,81,100} In the setting of combined ACL and PCL injury, reconstruction of the ACL prior to addressing PCL insufficiency will predictably lead to ACL failure.⁵⁸ As the medial meniscus is a secondary constraint to anterior tibial translation, the graft within a medial meniscus-deficient knee experiences increased forces that contribute to early failure.^{63,77} Meniscal transplant should be considered in these patients.^{11,21}

Primary Reconstruction Graft Choice

Numerous studies and meta-analyses of primary hamstring and patellar tendon autograft ACL reconstructions have demonstrated average failure rates of 3.6%, with no difference in failure rates between these autografts.^{10,34,137-142} However, allograft reconstructions have a three- to five-times higher likelihood of graft failure compared with autograft.^{48,101-103} This may be attributable, in

part, to patients having relatively less postoperative pain and a more rapid postoperative rehabilitation following allograft reconstruction, thereby returning to a higher activity level prior to adequate graft healing.^{34,104} Allograft processing may also increase the risk of graft failure, as irradiated grafts have demonstrated a significantly higher failure rate compared with nonirradiated grafts.^{103,105}

Concomitant Intra-Articular Pathology

Knees undergoing revision ACL reconstruction have a higher incidence of chondral injuries and meniscal tears compared to primary reconstruction.^{24,28,52,106,107} In one study, 90% of knees undergoing revision reconstruction had a meniscus or chondral injury and greater than 50% demonstrated both.²² Similarly, MARS group reported modified Outerbridge grade 2 or higher lesions in 73% of patients undergoing revision, with concurrent meniscal and cartilage damage in 57%.²²

Meniscus Injury

Although it is known that the medial meniscus plays a critical role in limiting anterior tibial translation in an ACL-deficient knee, it remains unknown if there is a critical amount of medial meniscus that, if removed, predisposes the reconstruction to failure.^{21,108} Previous partial meniscectomy, but not meniscal repair, is associated with a higher incidence of articular cartilage lesions at the time of revision.^{21,106} The MARS group demonstrated an overall 74% incidence of meniscal injury at the time of revision reconstruction,²² similar to that at the time of primary reconstruction.¹⁰⁹⁻¹¹¹ The prevalence of medial meniscus tears at the time of revision is 40% to 46%, higher than at the time of primary reconstruction.^{8,22,112} Conversely, there is actually a decreased incidence of new, untreated lateral meniscal tears at revision compared with primary reconstruction.¹¹ Thus patients undergoing revision due to recurrent instability appear to be continuing to put the medial meniscus at risk for further injury.

Articular Cartilage Injury

Inferior patient-reported outcome scores have been reported following revision ACL reconstruction compared with primary reconstruction, and this is associated with an increased incidence and severity of chondral lesions at the time of revision.^{6,53} Although it is unlikely that these lesions affect stability in the reconstructed knee, increased chondrosis at the time of revision may have detrimental effects on clinical outcomes despite appropriate surgical technique, graft healing, and adequate clinical laxity.²⁹ Even when controlling for meniscus status (prior meniscectomy versus no prior meniscectomy), there is an increased risk of lateral and patellofemoral compartment chondrosis at the time of revision compared with primary reconstruction.¹¹ There is a strong association between ACL deficiency and acceleration of degenerative changes,^{23,113,114} and the status of the articular cartilage at the time of revision reconstruction may be one of the most important predictors of a successful clinical outcome.³³ Patients undergoing delayed revision reconstruction (greater than 6 months following onset of symptomatic instability) have a markedly higher incidence of articular cartilage

degeneration compared with earlier revisions (53% vs. 24%), and the prevalence of advanced degenerative changes is also significantly higher in the delayed group.²³ Early restoration of stability may lead to reduced secondary articular cartilage damage and a return to previous activity levels,^{7,23,115} and it has been suggested that revision reconstruction should be performed within 6 months of graft failure to minimize the risk of degenerative changes and arthritis progression.^{23,53,116}

TREATMENT OPTIONS

Several factors must be considered in the setting of revision ACL reconstruction, including graft selection, incision locations, tunnel locations, tunnel sizes and need for bone grafting, previous fixation type and need for removal, method of revision fixation, and postoperative protocol. The revision reconstruction often must be adapted to the technique previously used.⁹¹ The revising surgeon must be proficient with a variety of techniques to address such issues as retained implants, malpositioned tunnels, bone loss, and expanded tunnels. Additional knee pathology, including meniscus or ligamentous injury, may be addressed at the time of revision ACL reconstruction or staged, with the revision ACL reconstruction often the final procedure. Although most revisions are performed as single-stage procedures, specific situations in which two-stage procedures should be considered are patients with incomplete ROM requiring lysis of adhesions, malalignment requiring corrective osteotomy, significant tunnel widening necessitating bone graft, and infection.⁹¹

Evaluation of Tunnels

Prior tunnel placement may be categorized in one of three ways: (1) accurate and not requiring redirection, (2) completely inaccurate in a location that will not interfere with new tunnel creation, or (3) overlapping, such that the prior tunnel and a properly placed tunnel will partially overlap.⁷⁶ Partially overlapping tunnels are generally the most challenging of these three because they require the surgeon to adjust his or her technique due to enlarged tunnels. The scope should be inserted into each tunnel to assess tunnel size and the quality of the surrounding bone to determine the need for grafting and the best method of fixation.⁹¹ The tibial tunnel can be viewed by placing the scope directly into the tunnel through the previously made tibial skin incision. The femoral tunnel can be visualized by placing the scope through the previous anteromedial portal.

Prior Fixation

Bioabsorbable implants are generally difficult to remove and may be left in place and overreamed after guide pin placement. However, metal interference screws must be removed if interfering with the planned tunnel location. In the rare case that properly placed tunnels can be made independent of prior tunnels and metallic implants, the implants may be left in place (Fig. 99.4). If metallic hardware is removed, the size and location of the resulting defect must be considered. The defect may necessitate grafting, either as part of a single-stage or two-stage revision, or the use of larger bone blocks with a patellar tendon autograft or various bone block allograft options.

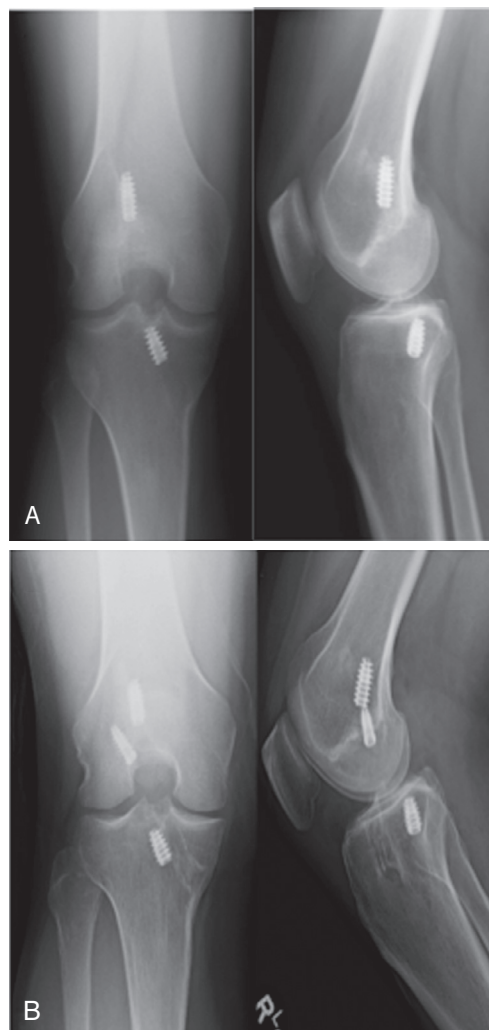


Fig. 99.4 Correction of femoral tunnel malpositioning. (A) Preoperative radiographs with fixation from prior failed anterior cruciate ligament (ACL) reconstruction demonstrate excessively posterior tibial tunnel and vertical femoral tunnel. (B) Postoperative radiographs following revision ACL reconstruction with new tibial tunnel anterior to Blumensaat line and less vertical femoral tunnel at approximately 1030 position.

Tunnel Grafting

Bone grafting of existing tunnels should be strongly considered when expanded tunnels measure more than 15 mm, because ACL graft options are limited and fixation may be compromised in tunnels this large.⁷⁶ Bone grafting may be performed at the time of revision ACL reconstruction or as a staged procedure, with staged reconstructions generally performed for larger defects (Fig. 99.5). Of all patients undergoing revision ACL reconstruction, approximately 9% undergo tibial tunnel grafting and 8% undergo femoral tunnel grafting as a staged procedure prior to revision reconstruction.²² Conversely, concomitant revision reconstruction and bone is performed in only 3% of patients.²² In the setting of a two-stage procedure, serial radiographs should be obtained to confirm complete graft incorporation prior to the second-stage procedure, which generally follows grafting by approximately 6 months.⁷⁶

Prior to grafting, a thorough débridement of any prior graft and fixation material must be performed to facilitate new graft

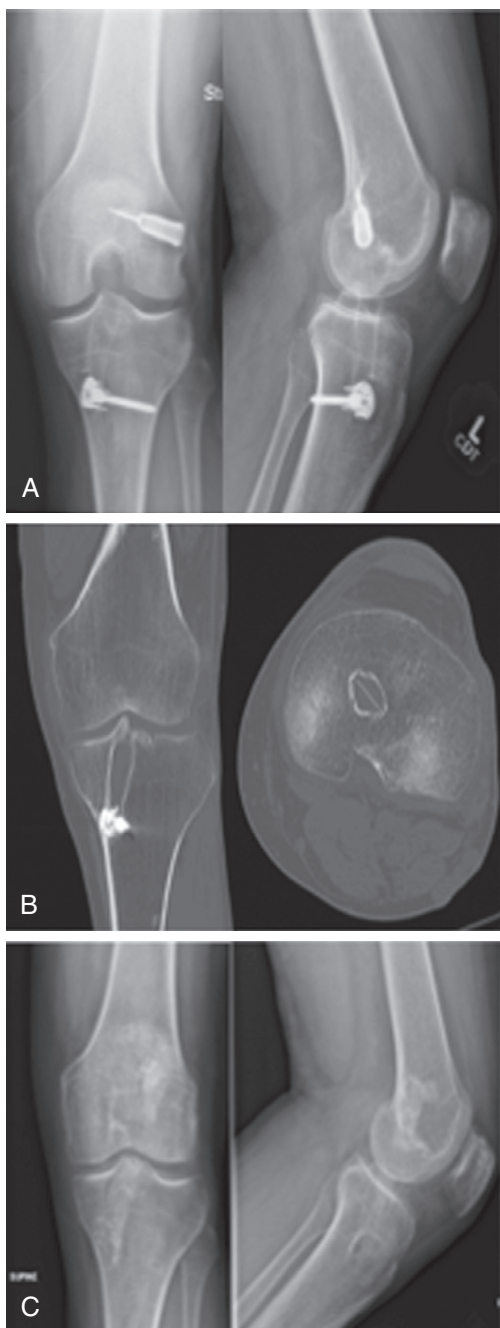


Fig. 99.5 Tunnel expansion in revision anterior cruciate ligament (ACL) reconstruction. (A) Preoperative radiographs prior to revision ACL reconstruction demonstrate fixation from prior failed ACL reconstruction. Tunnels are in a position that would overlap with planned revision ACL tunnels. (B) Computed tomography obtained prior to revision ACL reconstruction demonstrates expansion of tibial tunnel, with tunnel diameter measuring approximately 17 mm. Tunnel expansion would preclude a single stage procedure. (C) Revision ACL reconstruction was performed in staged manner, with allograft dowel plugs used to fill previous tunnels 6 months prior to revision ACL reconstruction.

healing to native bone. Autograft may be harvested from Gerdy tubercle, distal femur, or in the setting of larger defects, the iliac crest. Cancellous allograft can be used alone or in combination with autograft. Alternative autograft sources, including single or multiple press-fit osteochondral autograft transplantation (OAT)

plugs harvested from the iliac crest or medial tibial metaphysis,^{118,119} allograft dowel plug grafting (which may allow for larger diameter plugs), and calcium phosphate putties have been described as well, but no long-term results of biologic incorporation in the setting of revision ACL reconstruction are available at this time.¹²⁰ Expanded tibial tunnels may be filled with graft placed in an inferior to superior direction directly through the tibial skin incision using a small tamp. When grafting the femoral side, dry arthroscopy may be useful to prevent washing out graft. During femoral tunnel grafting, a small arthroscopic cannula or skid can be placed through an accessory medial portal made in line with the previous femoral tunnel. Allograft dowels may be sized to fit the dilated tunnel size and impacted into place.¹²¹

Tunnel Preparation

Revision ACL reconstruction is typically performed using a single-bundle technique with tunnels placed in the center of the anatomic footprints.⁷⁶ In the setting of overlapping tunnels, in which the previous and newly planned tunnels will overlap, the new femoral or tibial tunnels may be created using the concept of divergent tunnels.⁸⁷ Although the previous and new tunnels may converge as they approach the intracondylar notch, the tunnels should separate as they progress further away from the joint, and the new tunnel should sufficiently diverge from the previous tunnel to provide a cylinder of native bone for adequate graft fixation.⁷⁶ In the setting of tunnel overlap, an expanded tunnel may also be created by progressively reaming until a uniform cylinder results. Following this, depending on defect size, a custom autograft or allograft plug may be sized to fill the entire defect.^{83,122}

Femoral Tunnel

Transtibial drilling of the femoral tunnel limits the position from which the femoral footprint can be accessed. Other options that can be used to access the femoral footprint in an anatomic trajectory include outside-in femoral tunnel creation using a rear-entry technique or using an accessory anteromedial portal, both of which allow creation of an independent femoral tunnel. Notchplasty in the revision setting may result in inferior patient-reported outcomes and should not routinely be performed unless deemed necessary at the time of surgery.¹²³ For the outside-in rear-entry technique, an incision is made over the lateral aspect of the distal femur, splitting the iliotibial band to allow introduction of a drilling guide (Fig. 99.6). Care should be taken to avoid prior tunnels by using a more horizontal trajectory. Allowing for creation of a completely new femoral tunnel, this technique can provide an excellent way to obtain a new fixation point in the revision setting. Alternatively, an accessory anteromedial portal can be made using arthroscopic visualization with a spinal needle, as varying inferior and medial portal positioning provides a variety of angles to approach the femoral footprint. The knee should be hyperflexed to optimize the length and anterior direction of the newly created tunnel, thereby helping to prevent posterior condyle blowout and peroneal nerve injury.¹²⁴ A guide pin may then be placed through this newly created accessory medial portal, directed into the femoral footprint, and driven

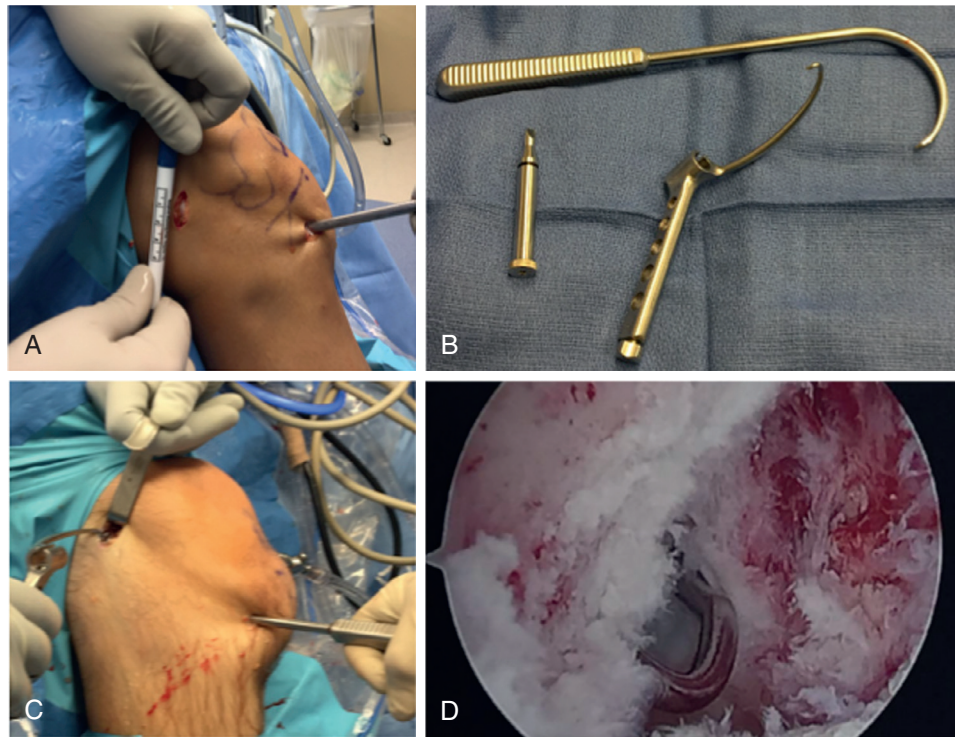


Fig. 99.6 Outside-in rear-entry femoral tunnel preparation. (A) Lateral thigh incision for outside-in rear-entry technique. (B) Guides for rear-entry femoral approach. (C) Rear entry guide pulled through anterolateral portal into notch with curved hook. (D) This technique allows the ability to place guide and drill tunnel at desired location within the intercondylar notch—here femoral tunnel is placed low in notch.

through the lateral cortex of the femur. The guide pin should exit above the level of the superior pole of the patella and anterior to the midline of the femur. With the knee remaining hyperflexed and the surgeon taking care to avoid iatrogenic articular cartilage damage, the guide pin should be overreamed to complete creation of the new femoral tunnel. Use of a unipolar or low-profile reamer may decrease the risk of iatrogenic articular cartilage damage. Following creation of the femoral tunnel, a motorized shaver may be used to remove debris.

Tibial Tunnel

Tibial tunnel management is generally less challenging than on the femoral side because the tibial tunnel is directly accessible through a prior open incision. A variable-angle drilling guide should be directed to enter the tibial articular surface in an anatomic and, if possible, a distinct location from the prior tunnel in order to facilitate adequate fixation. The angle of the new tunnel may be adjusted to create a longer tunnel, if necessary. An excessively posterior tibial tunnel presents a challenge in that the new, more anterior tunnel may overlap with the previous tunnel. Analogous to a blocking screw, a bioabsorbable screw or allograft bone dowel may be placed posteriorly to fill the previous tunnel and facilitate new graft placement in a more anterior position.

Graft Fixation

Prior to graft insertion and fixation, the new tunnel diameters, bone quality, and graft size must be carefully evaluated to ensure

an environment that is adequate for graft fixation and healing. If tunnel expansion is anticipated or occurs following tunnel creation, soft tissue grafts should be used with caution. If 2 to 3 mm of graft-tunnel mismatch occurs, a stacked-screw technique may be used.⁷⁶ Alternatively, a bone-patellar tendon-bone (BPTB) autograft with appropriately sized blocks to fill the mismatch may be used. In the setting of larger graft-tunnel mismatch, allograft with affixed bone block (Achilles tendon, BPTB) with appropriately large bone plugs should be prepared taking into account the mismatch. Synthetic dowels and allograft bone plugs have also been used to provide supplemental interference fixation within tunnels.^{83,122} In the setting of a loss of tunnel integrity, such as femoral tunnel blowout, or if the bone quality is poor, alternative methods of fixation besides inference screws should be considered. Laterally based bioabsorbable cross-fixation pins and arthroscopic buttons can provide improved fixation on the intact lateral cortex in this setting. Stay sutures may also be inserted through the graft and fixed to a femoral post, with a supplemental post placed on the lateral femur or anterior tibia. Recently, the MARS group has demonstrated improved results with metal screw fixation, and, although the reasons for these results are not yet understood, this should be taken under consideration.¹²³

Double-Bundle Reconstruction in Revision Setting

Although double-bundle reconstruction has gained popularity in the past decade,¹²⁵⁻¹²⁷ there are no absolute indications for this technique in the setting of primary or revision ACL

reconstruction.^{76,128,129} A failed double-bundle primary reconstruction may present specific challenges because prior tunnel placement with two separate tunnels and soft tissue grafts often interferes with planned tunnels at the time of revision. Although it has been described as feasible, there is currently no evidence that double-bundle reconstruction provides improved stability or clinical outcomes compared to single-bundle reconstruction in the revision setting.^{130,131} In addition, conversion from a single- to a double-bundle reconstruction may require staged procedures, with initial bone grafting followed by revision reconstruction, which may unnecessarily delay revision reconstruction, increasing the likelihood of progressive chondrosis.²³

In the setting of a vertical graft with an excessively posterior tibial tunnel that provides adequate sagittal plane stability but inadequate rotatory stability, an augmentation procedure may be considered in which the previous vertical graft is left intact.¹³¹ A second graft may be placed anterior to the excessively posterior tibial tunnel into a femoral tunnel site located within femoral footprint of the posterolateral bundle. No long-term outcomes following this procedure have been reported, and due to a lack of available data at this time, we cannot recommend the use of any double-bundle reconstruction technique in the revision setting.

Graft Choice

Multiple factors impact the choice of graft in revision ACL reconstruction, including age, sex, previous graft choice, ACL revision number, concurrent ligamentous repair, surgeon preference, and the surgeon's opinion of the prior failure.¹³² Although graft choice may impact outcomes in the revision setting, the ability to choose a graft may be limited by a number of other factors.

Factors Affecting Graft Choice

Graft choices for revision reconstruction include semitendinosus and gracilis (hamstring autograft), BPTB, quadriceps tendon, and various allograft choices.^{91,138} If autograft is chosen, it should be harvested only after determining that all of the technical steps of the revision reconstruction can be successfully performed.⁹¹ In 2010 the MARS group investigated graft choice at the time of revision ACL reconstruction and found that autograft was used in 45% of cases and allograft in 54%; 64% of autografts 55% of allograft were BPTB. Hamstrings graft made up 29% of autografts and only 5% of allografts.²² More recently, the MARS group reported that autograft was used in 48% of revision reconstructions and allograft in 49%, with 3% using a combination of both.¹³² Surgeon preference was found to be the most important factor in determining revision graft choice. Other factors affecting graft choice were prior graft type and patient age. If an autograft was used previously, an allograft was 3.6 times more likely to be chosen for the revision. It is well established that there is a higher rate of allograft failure in young, active athletes following primary reconstruction,³⁴ and recent data suggest higher rates of reoperation following prior reconstruction with allograft.¹³³ The surgeon's opinion of the failure has also been shown to impact graft choice at the time of revision, and a belief that an allograft failed for biologic reasons leads to a higher likelihood of using autograft.

Comparison Among Graft Choices

The use of autograft hamstring, BPTB, and combinations of hamstring, BPTB, and quadriceps tendon autograft has yielded results similar to those reported after primary reconstruction in terms of objective laxity.^{51,53,55,56,91,116} The MARS group reported improved patient-reported outcome scores and sports function following autograft revision reconstruction compared with allograft.¹³⁴ Furthermore, autograft resulted in a decreased risk of graft failure at 2-year follow-up, and these patients were 2.78 times less likely to undergo subsequent revision reconstruction compared with allograft. No studies have demonstrated differences in terms of graft failure or patient-reported outcomes comparing soft tissue and BPTB autografts. Key factors predisposing to allograft failure appear to be patient age, activity level, and the method of graft processing.^{34,135} Irradiation of allograft likely adversely affects graft laxity, predisposing to reconstructions failure.^{49,136}

In the setting of graft failure without evidence of technical error from the original procedure, the contralateral or ipsilateral patellar tendon may be the preferred graft source. The contralateral hamstring tendon may be considered as a graft choice if the primary procedure was performed with ipsilateral hamstring autograft.²⁹ Allograft should be given considering when tunnel expansion precludes the use of autograft alone.

Two-Stage Revision Anterior Cruciate Ligament Reconstruction

Two-stage revision reconstruction is often performed to avoid overlap between tunnels used at the time of revision ACL reconstruction and those made during the previous reconstruction.²⁴ Because increased time to revision correlates with a higher incidence of meniscal and chondral lesions,^{23,53} the revising surgeon must be judicious when deciding whether a two-stage procedure is necessary or a single-stage procedure will suffice. A two-stage procedure generally requires a 6-month window between procedures in which the patient may continue to have instability.²⁴ When possible, preference should be given to a single-stage procedure as long as appropriate tunnel placement and graft fixation can be achieved. A two-stage revision should be reserved for cases in which tunnel expansion precludes a single-stage procedure.

POSTOPERATIVE MANAGEMENT

Principles of Rehabilitation

The principles of postoperative management following revision ACL reconstruction are similar to primary reconstruction.¹⁴³ However, although specific protocols describing the goals and timing for rehabilitation following primary reconstruction are well described,¹⁴³⁻¹⁴⁵ there are limited data on rehabilitation following revision surgery. Rehabilitation and eventual return to sport are generally slower than primary reconstruction,¹⁴⁶ especially in the setting of allograft reconstruction in which graft incorporation may take longer.^{147,148} Overaggressive rehabilitation may result in early graft failure or late-onset laxity.²⁹ As previously mentioned, revision patients are more likely to have concomitant intra-articular pathology addressed at the time of



Authors' Preferred Technique

General Principles

As previously mentioned, surgical options may be limited in the revision setting by what was done during the previous reconstruction, as well as based on the etiology of graft failure. Prior graft selection may limit choices in the revision setting. The location and size of previous tunnels may dictate whether graft is necessary and whether the procedure may be successfully completed in one or two stages. The authors use a single-stage procedure unless tunnel dilation exceeds 15 mm, in which case the tunnels are grafted and the revision reconstruction performed in a staged fashion, typically approximately 6 months following tunnel grafting and complete graft incorporation. The second stage procedure is then performed similar to a primary ACL reconstruction but generally rehabilitated more slowly, as described later.

When possible, the authors use autograft, even if the contralateral extremity needs to be used. The authors use a rear-entry two-incision technique unless the locations of prior incisions prohibit this. A rear-entry two-incision approach allows femoral tunnel drilling to be performed independent of the tibial tunnel position, thereby allowing more flexibility in the revision setting. What remains most critical is creating tunnels in a fashion that the surgeon is comfortable with while allowing for adequate fixation. We describe our preferred technique for single-stage, single-bundle revision ACL reconstruction.

Diagnostic Arthroscopy

Standard diagnostic arthroscopy is performed, generally using prior portal incisions. Any additional intra-articular pathology may be treated prior to beginning revision ACL reconstruction. The remnant ACL is débrided and prior tunnel locations identified and assessed. The location of planned tunnels should be compared with the existing tunnels. Hardware is removed if it will affect the revision.

Tunnel Preparation

An incision over the planned tibial tunnel site is made, usually using or extending the prior incision used for tibial tunnel preparation. The exit point of the planned tibial tunnel should lie at the anatomic insertion of the native ACL. If the exit point of the existing tibial tunnel is appropriate, then the newly planned tunnel may be designed to converge with the existing tunnel at the articular surface. If this approach is used, care must be taken to ensure that the two tunnels diverge as they move away from the joint to prevent excessive tunnel dilation. Alternatively, if the previously drilled tunnel is in an appropriate location without excessive tunnel dilation, any hardware interfering with fixation may be removed and the same tunnel used.

Next, the femoral tunnel is prepared. The native femoral insertion should be identified and any graft stump débrided from this location. An accessory antero-medial tunnel is created or extended using a prior incision. Care must be taken not to violate the posterior condylar wall if the newly planned tunnel lies posterior to the previously drilled tunnel. If the previously drilled tunnel enters the joint appropriately at the native insertion, then converging tunnels may be used, as described previously for the tibia. Alternatively, if the previous drilled tunnel is in an appropriate, anatomic position, it may be used in a similar fashion as described for the tibia.

Graft Insertion

Following tunnel preparation, the graft is passed from the distal aspect of the tibial tunnel and pulled through the femoral tunnel similar to a primary reconstruction. The authors generally use interference screw fixation in both the primary and revision setting.

surgery, which can also slow rehabilitation depending on which specific structures are injured or repaired.^{84,148,149}

Rehabilitation Protocol

Phase 1: Acute Postoperative Phase (0 to 4 Weeks)

The patient is allowed weight bearing as tolerated beginning immediately postoperative as long as there are no contraindications due to other intra-articular injuries. Similar to previous findings in primary ACL reconstructions, recent data demonstrated no improvement from a rehabilitative brace or knee immobilizer (Wright et al., 2017, unpublished data). Crutches are provided for assistance. Patients begin quadriceps activation with active knee ROM and gentle assist on postoperative day 1, with a goal of at least 90 degrees of knee flexion and full extension. Elevation, compression, and icing should be used to minimize effusion.¹⁵⁰ Specific exercises in the acute postoperative phase include quadriceps sets, straight-leg raises, and hamstring activation (heel slides, standing knee flexion). Gait training may begin during the later stages of this first phase.¹⁴⁹ Closed kinetic chain (CKC) exercises, including heel raises and mini-squats, may also commence late in the first stage.

Phase 2: Subacute Postoperative Phase (2 to 8 Weeks)

As the knee joint effusion and pain levels decrease over the first postoperative month, the ability to weight bear without assist and achieve greater motion will also improve. Exercises to improve motion including supine wall slides, active-assisted

heel slides, and prone/standing hamstring curls.¹⁴⁹ Aggressive passive knee flexion should be avoided. Patients who have advanced through the first phase may progress to limited open kinetic chain (OKC) exercise (limited isometric knee extension from 45 to 90 degrees), additional CKC exercises through a greater ROM (terminal extension with TheraBand, split squat, hip dominant squat, leg press, step-ups), core strengthening, more advanced gait training, balance exercises, and low-level cardiovascular conditioning including stationary bike.^{149,151-154} Aquatic therapy may also be implemented.

Phase 3: Neuromuscular Conditioning (6 Weeks to 8 Months)

Knee ROM should be full extension to greater than 120 degrees of flexion and a normal gait achieved by approximately 12-weeks postoperative. Strengthening and cardiovascular exercises may be intensified. Until approximately 3 months postoperative, CKC exercises predominate, including body-weight squatting, lunges, and leg press.

At 3 months postoperative, patients should not yet be progressed to agility training, instead focusing on an increased intensity of both OKC and CKC exercises, along with progression of core exercises. At approximately 5 to 6 months postoperative, complex movement training reflecting the patient's demands should be implemented. To begin these exercises, which often impart rotational forces about the knee, the patient must have sufficiently progressed in terms of strength and ROM. Cardiovascular training

may include progression to jogging and careful incorporation of form running drills, with change of direction drills introduced late in the phase as indicated by patient progression.¹⁴⁹

Phase 4: Return to Activity (8 Months to 1 Year)

Critical assessment of patient-specific demands in the context of rehabilitation milestones must take place prior to return to full activity. Requisite neuromuscular control and endurance must be achieved to safely meet their sport/activity-specific demands. Functional testing may be performed, including single-hop tests, vertical leap, agility tests, and functional performance testing.^{149,155} Reaction drills may be also implemented.¹⁵⁶ Once all deficiencies have been addressed and the patient is psychologically prepared for return to preinjury level of activity, they may be cleared for full a return.^{157,158} Recent data suggest that the use of an ACL derotation brace at the time of return to sport may improve patient-reported outcome scores among patients undergoing revision reconstruction and thus should be considered (Wright et al., unpublished data 2016).

RESULTS

Results following revision ACL have been repeatedly demonstrated as inferior to those after primary reconstruction. Although objective rotatory and sagittal laxity can be restored to similar levels as those seen after primary reconstruction, failure rates following revision are three to four times higher than after primary reconstruction.⁹ Clinical outcomes scores are also significantly worse compared with primary reconstruction, with lower Cincinnati, Lysholm, Tegner, Marx activity, IKDC, and multiple Knee injury and Osteoarthritis Outcomes Score (KOOS) subscale scores.^{9,28,76,159} However, these measures are all significantly improved compared with before revision reconstruction.^{3,6,29,87} Approximately 40% of patients undergoing revision ACL reconstruction do not return to the same level of sport or competition.^{6,33,54,57,91,98,116,160}

An increased incidence of meniscal and chondral injuries plays significant role in the inferior outcomes compared with primary reconstruction.^{6,53} Multiple studies have demonstrated inferior outcomes in patients undergoing revision reconstruction when chondral lesions or meniscal tears are present at the time of revision.^{6,53} One study found lower subjective and objective IKDC scores in patients who had undergone prior meniscectomy or were found to have significant cartilage damage at the time of revision ACL reconstruction.⁶

Multiple Revisions

Repeat revision ACL reconstruction restores AP and rotational graft stability while improving the functional outcomes in patients who have failed prior revision reconstruction. However, most patients do not return to the same level of activity following repeat revision.^{3,21,117} Although the most common cause of first-time recurrent laxity is a traumatic noncontact injury, multiple-revision patients tend to have recurrent instability without a traumatic injury.²¹ This suggests that there may be fundamental differences in how ACL grafts fail in the multiple-revision setting. There is also an increased rate of chondral injuries both within the medial and patellofemoral compartments, suggesting that

medial compartment degeneration may be an important factor in recurrent graft injury.²¹

There have been limited reports of multiple revisions, with failure rates ranging from 0% to 13%.^{3,117} Repeat revision for recurrent instability without an identifiable traumatic event, the presence of International Cartilage Repair Society (ICRS) grade 3 or 4 chondrosis, and obesity have been associated with inferior outcomes following multiple revision reconstruction.^{3,117} In one study, technical error (most often tunnel malposition) was more common in the multiple-revision group than the first-time revision group. Furthermore, staged bone grafting is performed 22% of the time in the multiple-revision knee, double the rate of first-time revision, suggesting that the remaining host tibial and femoral bone is limited after two or more ACL reconstructions. These data also suggest that femoral tunnel malpositioning should be corrected at the time of first revision to lower the risk of recurrent failure.

COMPLICATIONS

Revision ACL reconstruction has a higher complication rate than primary reconstruction, but overall numbers are still relatively low.^{7,21,29,61,161} The types of complications following revision reconstruction are similar to those following primary surgery, including infection, stiffness, and venous thromboembolism. Deep surgical site infections occur in 0.5% to 0.8% of patients undergoing revision reconstruction and venous thromboembolism in less than 0.5%.^{21,162}

FUTURE CONSIDERATIONS

Revision remains a challenging clinical scenario that requires the widest skill set of any sports procedure including hardware removal, bone grafting, and meniscal and chondral work, along with consideration of alignment and graft choices. The challenge remains to return patients to their previous activity and prevent future injury. Continued evaluation by large prospective cohorts will hopefully continue to elucidate subtle issues contributing to failure and worse outcomes. The MARS group embarked upon a 10-year follow-up with onsite evaluation that may show new findings not found with questionnaire-based follow-up.

For a complete list of references, go to [ExpertConsult.com](https://www.expertconsult.com).

SELECTED READING

Citation:

Wright RW, Gill CS, Chen L, et al. Outcome of revision anterior cruciate ligament reconstruction: a systematic review. *J Bone Joint Surg Am.* 2012;94(6):531–536.

Level of Evidence:

IV, systematic review of level I-IV studies

Summary:

This systematic review evaluates 21 eligible studies reporting outcomes of revision ACL reconstruction at minimum 2-year follow-up. The authors report that ACL reconstruction results in a worse outcome compared with primary ACL reconstruction and a dramatically elevated failure rate, 3–4 times higher than primary ACL reconstruction.

Citation:

Group MARS, Wright RW, Huston LJ, et al. Descriptive epidemiology of the Multicenter ACL Revision Study (MARS) cohort. *Am J Sports Med.* 2010;38(10):1979–1986.

Level of Evidence:

II, cross-sectional study

Summary:

This study describes the formation of the prospective MARS cohort and reports patient demographic and clinical features of this group of patients. This comprises the largest revision ACL reconstruction cohort in the literature.

Citation:

MARS Group. Effect of graft choice on the outcome of revision anterior cruciate ligament reconstruction in the Multicenter ACL Revision Study (MARS) cohort. *Am J Sports Med.* 2014;42(10):2301–2310.

Level of Evidence:

II, prospective cohort study

Summary:

This multicenter prospective study of 1205 patients demonstrates improved sports function and patient-reported outcomes decreased risk of graft rerupture after use of autograft compared with allograft in revision ACL reconstruction.

Citation:

Borchers JR, Kaeding CC, Pedroza AD, et al. Intra-articular findings in primary and revision anterior cruciate ligament reconstruction surgery: a comparison of the MOON and MARS study groups. *Am J Sports Med.* 2011;39(9):1889–1893.

Level of Evidence:

II, prospective cohort study

Summary:

This study compares meniscal and articular cartilage injuries found at the time of primary and revision ACL reconstruction using two of the largest available prospectively collected patient cohorts. The authors report a significantly higher incidence of lateral and patellofemoral compartment chondrosis in the revision setting, with a higher incidence of articular cartilage damage following prior meniscectomy.

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